HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed precise technical drawings for solar PV systems, including rooftop and ground-mounted installations, ensuring compliance with Irish and EU standards.
- Interpreted site survey data and translated it into accurate design plans, preparing drawings, bill of materials (BOM), and wiring diagrams for domestic properties.
- Collaborated with sales, project management, and installation teams to ensure project delivery within timelines and budget constraints.
- Designed and implemented Python scripts to automate data extraction from site survey reports, reducing manual data entry time by approximately **30%**.
- Engineered automated report generation for project documentation using Python, consolidating BOM and wiring diagram data into structured outputs.
- Managed production environments for enterprise applications, responding to P1/P2 incidents and maintaining service level agreements.
- Containerized design tools and utilities using **Docker** for consistent deployment across development and project teams.

SKILLS

Design Software – AutoCAD, Solar PV designing tools, Helioscope, PVSyst, PVSol, HOMER

Engineering – Solar PV Design, Electrical Engineering, Renewable Energy, Technical Drawings, Bill of Materials (BOM), Wiring Diagrams

Programming – Python, Java, SQL

Cloud & DevOps – AWS, Docker, Jenkins, Git

Data Analysis & BI – Power BI, DAX, Power Query, data modelling

Methodologies – Agile, Scrum

PROJECTS

Automated Solar PV Design Documentation

- Developed a Python-based tool to parse site survey data and automatically generate preliminary design plans and BOMs for domestic solar PV installations.
- Integrated with **AutoCAD** scripting capabilities to pre-populate technical drawing templates, reducing manual drafting time by an estimated **40%**.
- Designed a modular data processing pipeline to accommodate varying input formats from different survey methods, ensuring broad applicability.

Renewable Energy Site Survey Data Analysis Dashboard

- Built an interactive **Power BI** dashboard to visualize key metrics from solar PV site surveys, including roof space, shading analysis, and potential energy yield.
- Created complex **DAX** measures to calculate estimated system performance and ROI based on survey inputs, facilitating informed sales discussions.
- Developed data transformation scripts in **Power Query** to clean and standardize diverse survey data sources before loading into the Power BI data model.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)